

No. of Functional Features Included in the Solution

Date	09 July 2026
Team ID	SL-2026-02
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Applicant Data Input	Allows users to enter loan applicant details through a web form.	User Interface	Done	High
2	Data Preprocessing	Cleans, validates, encodes, and transforms user input before prediction.	Data Processing	Done	High
3	Machine Learning Model	Predicts loan eligibility using the trained machine learning model.	Prediction Engine	Done	High
4	Loan Eligibility Prediction	Displays whether the applicant is eligible for a loan based on model output.	Prediction Module	Done	High
5	Flask Web Application	Connects the frontend with the machine learning model for real-time predictions.	Backend	Done	High
6	Responsive User Interface	Provides an easy-to-use interface across desktop and mobile browsers.	Frontend	Done	Medium
7	Model Loading	Loads the trained machine learning model and preprocessing objects using Joblib.	Backend	Done	Medium

8	Error Handling & Input Validation	Prevents invalid inputs and handles prediction errors gracefully.	Backend	Done	Medium
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Applicant Data Input, Data Preprocessing, Machine Learning Prediction, Flask Integration, Loan Eligibility Result
Metric	Count / Value
Total Features Planned	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Details Form, Loan Eligibility Result Page
2	Backend / Logic	4	Flask Routing, Data Preprocessing, Model Prediction, Error Handling
3	Database / Storage	2	CSV Dataset, Trained Model Files (.pkl / Joblib)
4	API / Integration	1	Flask Application Integration with Machine Learning Model
5	Security / Authentication	1	Input Validation and Secure Request Handling